
Kolmogorov equations for stochastic PDE's with multiplicative noise

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1 Introduction

We are here concerned with the following stochastic differential equation in the Hilbert space $H = L^2(0, 1)$,

$$\begin{cases} dX(t, \xi) = D_\xi^2 X(t, \xi) dt + g(X(t, \xi)) dW(t, \xi), & t \geq 0, \xi \in [0, 1], \\ X(t, 0) = X(t, 1) = 0, & t \geq 0, \\ X(0, \xi) = x(\xi), & x \in H, \xi \in [0, 1], \end{cases} \quad (1)$$

where g is a real function of class C^2 bounded together with its derivatives of order less or equal to 2 and W is a cylindrical Wiener process in H (see below for a precise definition). Existence and uniqueness of a solution of (1) are well known, see [16], [7]. Let us denote by R_t the corresponding transition semigroup,

$$R_t \varphi(x) = \mathbb{E}[\varphi(X(t, x))], \quad t \geq 0, x \in H, \quad (2)$$

where φ is a real bounded Borel function and $X(t, x)$ is the solution of equation (1).

The Kolmogorov equation corresponding to (1) reads as follows,

$$\begin{cases} u_t(t, x) = \frac{1}{2} \text{Tr} [\sigma^2(x) u_{xx}(t, x)] + \langle Ax, u_x(t, x) \rangle, & t \geq 0, x \in D(A), \\ u(0, x) = \varphi(x), & x \in H, \end{cases} \quad (3)$$

where A is the linear operator,

$$Ax = D_\xi^2, \quad x \in D(A) = H^2(0, 1) \cap H_0^1(0, 1),$$

and for any $x \in L^2(0, 1)$ the symmetric Nemitskii operator $\sigma(x) \in L(L^2(0, 1))$ is defined by

$$[\sigma(x)y](\xi) = g(x(\xi))y(\xi), \quad y \in L^2(0,1), \quad \xi \in [0,1]. \quad (4)$$

There is an increasing interest on infinite dimensional Kolmogorov equations, see the monographs [15], [5], [13], [8] (and references therein) and the papers [1],[17], [2], [3].

In particular, in [13] the case when the noise is additive is mainly considered with the exception of chapters 6 and 7. More precisely, Chapter 6 is devoted to Hölder continuous perturbations of the infinite dimensional Heat semigroup, see also some recent developpements in this direction in [2] and [3]. In chapter 7 of [13] the case when coefficients are of class C^3 is considered. Notice that in equation (1) the multiplicative noise can be written as $\sigma(x)dW(t)$ where for any $x \in L^2(0,1)$ the operator $\sigma(x) \in L(L^2(0,1))$ is operator defined by (4). Thus, in spite of the fact that g is C^2 , the mapping $\sigma(x)$ is only once Gateaux differentiable (except when g is constant). So, the method in [13, Chapter 7] does not work and some new technique has to be used.

The first main result of the paper is that if $\varphi \in C_b^3(H)$ there is a smooth solution of (3), see Theorem 13. It is well known that a candidate for the solution of (3) is given by

$$u(t, x) := R_t \varphi(x) = \mathbb{E}[\varphi(X(t, x))], \quad (5)$$

where R_t is the transition semigroup defined by (2). So, the proof of existence of a solution of (3) will consists in showing (by justifying the chain rule) that $u(t, x)$ is twice differentiable, that the trace of $\sigma^2 D^2 u(t, x)$ is finite and that equation (3) is fulfilled. However, these computations are not straightforward since σ is neither regular nor of trace class. This idea was applied in the case of reaction–diffusion equations with additive noise, see [6] and in the completely different situation of Navier–Stokes equations with additive noise, see [9].

We notice that the regularity we get for the solution of (3) is not enough to apply the Itô formula and so, to prove the uniqueness of a smooth solution of (3). This would require an additional job which we plan to make in a future paper.

In the second part of the paper we assume that $1/g$ is bounded so that there is a unique invariant measure μ , see [16]. We study here the Kolmogorov equation (2) in the space $L^2(H, \mu)$. It is well known that the transition semigroup R_t can be uniquely extended to a strongly continuous semigroup of contractions in $L^2(H, \mu)$. We shall still denote by R_t this extension and by L_μ the infinitesimal generator of R_t .

As a second main result of the paper we construct a core Γ for the generator L_μ consisting of regular functions and show that on Γ the operator L_μ is in fact a differential operator given by

$$L_\mu \varphi(x) = \frac{1}{2} \operatorname{Tr} [\sigma^2(x) D^2 \varphi(x)] + \langle Ax, D\varphi(x) \rangle, \quad x \in D(A), \quad \varphi \in \Gamma.$$

As it was pointed out in several previous situations, see e.g. [8] and references therein, to have an explicit expression of L_μ on the core Γ allows to prove easily the so-called “identité du carré des champs”,

$$\int_H L_\mu \varphi \varphi d\mu = -\frac{1}{2} \int_H |\sigma(x) D\varphi|^2 d\mu, \quad \varphi \in D(L_\mu). \quad (6)$$

Using (6) it is possible to prove that the derivative operator is closable in $L^2(H, \mu)$. This allows to define the Sobolev space $W^{1,2}(H, \mu)$ and to show that the domain of L_μ is included in $W^{1,2}(H, \mu)$.

Finally, we prove the Poincaré inequality, by generalizing a result proved in [10] in the case of additive noise. As a standard consequence we obtain the spectral gap of L_μ and the exponential convergence to equilibrium of R_t .

It would be interesting to consider the more general problem,

$$\begin{cases} dX(t, \xi) = (D_\xi^2 X(t, \xi) + f(X(t, \xi)))dt + g(X(t, \xi))dW(t, \xi), & \xi \in [0, 1], \\ X(t, 0) = X(t, 1) = 0, & t \geq 0 \\ X(0, \xi) = x(\xi), & x \in H, \xi \in [0, 1], \end{cases} \quad (7)$$

where f is a suitable real function. However, problem (7) does not seem to be a straightforward generalization of (1). It will be the object of a future research.

1.1 Notations

We denote by H the Hilbert space $L^2(0, 1)$ (norm $|\cdot|$, inner product $\langle \cdot, \cdot \rangle$). When there is the danger of confusion between the norm and the absolute value of a function x we shall write $|x|_{L^2(0,1)}$ instead of $|x|$. Moreover $L(H)$ (norm $\|\cdot\|$) will represent the Banach algebra of all linear bounded operators in H and $L_1(H)$ (norm $\|\cdot\|_{L_1(H)}$) the space of all trace class operators in H . We recall that

$$\|T\| = \sup\{|Tx| : x \in H, |x| = 1\}, \quad T \in L(H).$$

For any Hilbert space K (norm $|\cdot|$, inner product $\langle \cdot, \cdot \rangle$), we denote by $C_b(H; K)$ the linear space of all continuous and bounded mappings $\varphi: H \rightarrow K$. $C_b(H; K)$ endowed with the norm

$$\|\varphi\|_0 = \sup_{x \in K} |\varphi(x)|, \quad \varphi \in C_b(H; K),$$

is a Banach space.

Moreover $C_b^1(H; K)$ will represent the subspace of $C_b(H; K)$ of all functions $\varphi: H \rightarrow K$ which are Fréchet differentiable on H with a continuous and bounded derivative $D\varphi$. The space $C_b^k(H; K)$ for $k \geq 2$ are defined analogously. We shall write $C_b^i(H; \mathbb{R}) = C_b^i(H)$, $i \in \mathbb{N}$.

If $\varphi \in C_b^1(H)$ and $x \in H$, we shall identify $D\varphi(x)$ with the unique element h of H such that

$$D\varphi(x)y = \langle h, y \rangle, \quad x, y \in H.$$

If $\varphi \in C_b^2(H)$ and $x \in H$, we shall identify $D^2\varphi(x)$ with the unique linear operator $T \in L(H)$ such that

$$D\varphi(x)(y, z) = \langle Ty, z \rangle, \quad x, y, z \in H.$$

1.2 An extension of Gronwall's lemma

The following result is a generalization of a well known result.

Lemma 1. *Assume that $f: [0, +\infty) \rightarrow [0, +\infty)$ fulfills the inequality,*

$$f(t) \leq a(t) + b \int_0^t (t-s)^{-1/2} f(s) ds, \quad t \geq 0, \quad (8)$$

where a is continuous nonnegative and b is a nonnegative constant. Then we have,

$$\begin{aligned} f(t) &\leq a(t) + b \int_0^t (t-s)^{-1/2} a(s) ds \\ &\quad + \int_0^t e^{(t-s)\pi b^2} \left[a(s) + b \int_0^s (s-\sigma)^{-1/2} a(\sigma) d\sigma \right] ds. \end{aligned} \quad (9)$$

If, in particular, $a(t) = a$ we have

$$f(t) \leq ae^{\pi b^2 t} + 2ab \int_0^t s^{-1/2} e^{\pi b^2 (t-s)} ds \quad t \geq 0. \quad (10)$$

and

$$f(t) \leq 3ae^{\pi b^2 t} \quad t \geq 0. \quad (11)$$

Proof. We write (8) as

$$f \leq a + b\psi_{-1/2} * f, \quad (12)$$

where $\psi_{-1/2}(t) = t^{-1/2}$ and $*$ denotes the convolution ⁽¹⁾. Taking the convolution of both sides of (10) with $\psi_{-1/2}$ and taking into account that

$$(\psi_{-1/2} * \psi_{-1/2})(t) = \int_0^t (t-s)^{-1/2} s^{-1/2} ds = \pi,$$

yields

$$\psi_{-1/2} * f \leq a * \psi_{-1/2} + \pi b(1 * f). \quad (13)$$

Substituting this in (12) yields

¹ $(f * g)(t) = \int_0^t f(t-s)g(s)ds.$

$$f \leq a + b(a * \psi_{-1/2}) + \pi b^2(1 * f), \quad (14)$$

which is equivalent to

$$f(t) \leq a(t) + b \int_0^t (t-s)^{-1/2} a(s) ds + \pi b^2 \int_0^t f(s) ds. \quad (15)$$

Consequently

$$\begin{aligned} f(t) \leq a(t) + b \int_0^t (t-s)^{-1/2} a(s) ds \\ + \int_0^t e^{(t-s)\pi b^2} \left[a(s) + b \int_0^s (s-\sigma)^{-1/2} a(\sigma) d\sigma \right] \end{aligned}$$

Now (9) follows from the classical Gronwall lemma. Finally, (10) is clear and by (10) we have

$$\begin{aligned} f(t) \leq a e^{\pi b^2 t} \left(1 + 2b \int_0^t s^{-1/2} e^{-\pi b^2 s} ds \right) \\ \leq a e^{\pi b^2 t} \left(1 + 2b \int_0^\infty s^{-1/2} e^{-\pi b^2 s} ds \right), \end{aligned}$$

which yields (11). $\square$

2 Existence and uniqueness of solutions

2.1 The abstract setting

Let us write problem (1) in an abstract form introducing the linear self-adjoint operator $A: D(A) \subset H \rightarrow H$,

$$\begin{cases} Ax = D_\xi^2 x, & x \in D(A), \\ D(A) = H^2(0,1) \cap H_0^1(0,1), \end{cases}$$

where $H^i(0,1)$, $i = 1, 2$ denote the usual Sobolev spaces and

$$H_0^1(0,1) = \{x \in H_0^1(0,1) : x(0) = x(1) = 0\}.$$

We define moreover the (generally) nonlinear operator $\sigma: H \rightarrow L(H)$ by setting,

$$[\sigma(x)y](\xi) = g(x(\xi))y(\xi), \quad \xi \in [0,1], \quad x, y \in H.$$

We denote by (e_k) the complete orthonormal system in H consisting of the eigenfunctions of A ,

$$e_k(\xi) = \sqrt{\frac{2}{\pi}} \sin k\pi\xi, \quad \xi \in [0,1], \quad k \in \mathbb{N},$$

so that

$$Ae_k = -k^2\pi^2 e_k, \quad k \in \mathbb{N}.$$

Notice that

$$\|e^{tA}\| \leq e^{-\pi^2 t}, \quad t \geq 0.$$

Finally, we introduce the cylindrical white noise,

$$W(t) = \sum_{k=1}^{\infty} e_k \beta_k(t), \quad t \geq 0, \quad (16)$$

where (β_k) is a sequence of mutually independent standard Brownian motions on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$.

Now we can write problem (1) as follows,

$$\begin{cases} dX = AX dt + \sigma(X) dW(t) = AX dt + \sum_{k=1}^{\infty} [g(X)e_k] d\beta_k(t), \\ X(0) = x. \end{cases} \quad (17)$$

We shall solve equation (17) in the space $C_W([0, T], H)$ of all mean square continuous adapted (to the filtration $(\mathcal{F})_{t \geq 0}$) stochastic process $X(\cdot)$ defined in $[0, T]$ and taking values in H . It is well known that $C_W([0, T], H)$, endowed with the norm

$$\|X\|_{C_W([0, T], H)} = \left(\sup_{t \in [0, T]} \mathbb{E}(|X(t)|^2) \right)^{1/2},$$

is a Banach space.

Definition 2. A mild solution of equation (17) is a process $X \in C_W([0, T], H)$ such that

$$X(t) = e^{tA}x + \int_0^t e^{(t-s)A} \sigma(X(s)) dW(s), \quad t \geq 0, \quad x \in H. \quad (18)$$

In the following we shall denote by $X(\cdot, x)$ the solution of (18).

An important rôle will be played by the *stochastic convolution*,

$$W_X(t) = \int_0^t e^{(t-s)A} \sigma(X(s)) dW(s) = \sum_{k=1}^{\infty} \int_0^t e^{(t-s)A} [g(X(s))e_k] d\beta_k(s),$$

where $X \in C_W([0, T], H)$.

As we shall see, though the cylindrical white noise (16) does not leave in H (see e.g. [11, §4.3.1]), the stochastic convolution $W_X(t)$ does.

In order to study basic properties of $W_X(t)$ it is useful to introduce the function

$$F(t) = \sum_{h=1}^{\infty} e^{-th^2}, \quad t > 0. \quad (19)$$

Notice that

$$\begin{aligned} F(t) &\leq e^{-t} + \int_1^{\infty} e^{-tx^2} dx = e^{-t} \left(1 + \int_1^{\infty} e^{-t(x^2-1)} dx \right) \\ &= e^{-t} \left(1 + \int_0^{\infty} e^{-ty^2} \frac{y}{\sqrt{y^2+1}} dy \right) \leq e^{-t} \left(1 + \int_0^{\infty} e^{-ty^2} dy \right). \end{aligned}$$

So,

$$F(t) \leq e^{-t} \left(1 + 2t^{-1/2} \right) \leq 4t^{-1/2} e^{-t/2}, \quad t > 0. \quad (20)$$

Lemma 3. *Let $X \in C_W([0, T], H)$. Then we have*

$$\mathbb{E}(|W_X(t)|^2) = \int_0^t \sum_{h=1}^{\infty} e^{-2\pi^2 h^2 (t-s)} \mathbb{E}(|g(X(s))e_h|^2) ds. \quad (21)$$

Proof. Taking into account the independence of the (β_k) we have,

$$\mathbb{E}(|W_X(t)|^2) = \sum_{k=1}^{\infty} \int_0^t \mathbb{E}(|e^{(t-s)A}[g(X(s))e_k]|^2) ds.$$

Using the Parseval identity we find,

$$\begin{aligned} \sum_{k=1}^{\infty} |e^{(t-s)A}[g(X(s))e_k]|^2 &= \sum_{h,k=1}^{\infty} |\langle e^{(t-s)A}[g(X(s))e_k], e_h \rangle|^2 \\ &= \sum_{h,k=1}^{\infty} e^{-2(t-s)\pi^2 h^2} |\langle g(X(s))e_k, e_h \rangle|^2 = \sum_{h,k=1}^{\infty} e^{-2(t-s)\pi^2 h^2} |\langle e_k, g(X(s))e_h \rangle|^2 \\ &= \sum_{h=1}^{\infty} e^{-2(t-s)\pi^2 h^2} |\sigma(X(s))e_h|^2. \end{aligned}$$

So, (21) follows. $\square$

Proposition 4. *Let $X \in C_W([0, T], H)$. Then we have*

$$\mathbb{E}(|W_X(t)|^2) \leq 8\sqrt{\pi} \|g\|_0^2, \quad t \geq 0. \quad (22)$$

Moreover, for all $X, Y \in C_W([0, T], H)$ we have,

$$\begin{aligned} &\mathbb{E}(|W_X(t) - W_Y(t)|^2) \\ &= \frac{8\|g\|_1^2}{\pi\sqrt{2\pi}} \int_0^t e^{-(t-s)\pi^2} (t-s)^{-1/2} \mathbb{E}[|X(s) - Y(s)|^2] ds. \end{aligned} \quad (23)$$

Proof. By (21) we have, recalling that $|e_k(\xi)|^2 \leq \frac{2}{\pi}$ and taking into account (20),

$$\mathbb{E}|W_X(t)|^2 \leq \frac{2\|g\|_0^2}{\pi} \int_0^t F(2\pi^2 s) ds \leq \frac{8\|g\|_0^2}{\pi} \int_0^\infty e^{-\pi^2 s} s^{-1/2} ds$$

and (22) follows. The proof of (23) is similar. $\square$

2.2 Existence and uniqueness

The following result is well known, see e.g. [16], we present however the short proof for the reader's convenience.

Proposition 5. *For any $x \in H$ there exists a unique solution $X(\cdot, x)$ of equation (18).*

Proof. Write equation (18) in the form

$$X = e^{tA}x + \Lambda(X), \quad X \in C_W([0, T], H),$$

where

$$\Lambda(X)(t) = W_X(t), \quad t \in [0, T].$$

Then by (22) it follows that Λ maps $C_W([0, T], H)$ in itself. Let moreover $X, Y \in C_W([0, T], H)$. Then by (23) it follows that,

$$\begin{aligned} |\Lambda(X)(t) - \Lambda(Y)(t)| &\leq \frac{8\|g\|_1^2}{\pi\sqrt{\pi}} \int_0^t e^{-(t-s)\pi^2} (t-s)^{-1/2} ds \|X - Y\|_{C_W([0, T], H)} \\ &\leq \frac{16\|g\|_1^2}{\pi\sqrt{2\pi}} t^{1/2} \|X - Y\|_{C_W([0, T], H)}. \end{aligned}$$

Now let $T_1 \in (0, T]$ be such that $\frac{16\|g\|_1^2}{\pi\sqrt{2\pi}} T_1^{1/2} < 1$. Then Γ is a contraction on $C_W([0, T_1], H)$. Therefore, equation (18) has a unique solution on $[0, T_1]$. By a similar argument, one can show existence and uniqueness on $[T_1, 2T_1]$ and so on. $\square$

2.3 Galerkin approximations

It is useful to consider Galerkin approximations of equation (18). For any $n \in \mathbb{N}$ we denote by P_n the projector

$$P_n x = \sum_{k=1}^n \langle x, e_k \rangle e_k, \quad x \in H$$

and set $A_n = AP_n$. Then we consider the equation

$$X^n(t, x) = e^{tA_n}x + \sum_{k=1}^n \int_0^t e^{(t-s)A_n} [g(X^n(s, x))e_k] d\beta_k(s). \quad (24)$$

The following result is standard.

Proposition 6. *For any $T > 0$, $x \in H$ and $n \in \mathbb{N}$, there exists a unique solution $X^n(\cdot, x)$ of equation (24). Moreover,*

$$\lim_{n \rightarrow \infty} X^n(\cdot, x) = X(\cdot, x), \quad \text{in } C_W([0, T], H), \quad (25)$$

where $X(\cdot, x)$ is the solution of (18).

3 Kolmogorov equation

3.1 Setting of the problem

We are here concerned with the following *Kolmogorov equation*,

$$\begin{cases} u_t(t, x) = \frac{1}{2} \text{Tr} [\sigma^2(x)u_{xx}(t, x)] + \langle Ax, u_x(t, x) \rangle, & t \geq 0, x \in D(A), \\ u(0, x) = \varphi(x), & x \in H. \end{cases} \quad (26)$$

We are going to show that when the initial datum φ is sufficiently regular equation (26) has a solution in a classical sense. As it is well known, a candidate for the solution $u(t, x)$ of (26) is provided by the formula

$$u(t, x) = \mathbb{E}[\varphi(X(t, x))], \quad \varphi \in C_b(H), \quad t \geq 0, x \in H, \quad (27)$$

where $X(t, x)$ is the solution of (18). We shall check that, under suitable assumptions on φ , formula (27) produces in fact a solution of (26).

We shall need to consider the approximating equation

$$\begin{cases} u_t^n(t, x) = \frac{1}{2} \text{Tr} [P_n \sigma^2(x)u_{xx}^n(t, x)] + \langle A_n x, u_x^n(t, x) \rangle \\ u^n(0, x) = \varphi(P_n x), \end{cases} \quad (28)$$

which has a unique strict solution which we denote by $u^n(t, x)$.

3.2 Estimates for derivatives of $X(t, x)$

This subsection is devoted to establish some estimates concerning the derivatives X_x and X_{xx} , which will be used later. We start from the directional derivative

$$\eta^z(t, x) := X_x(t, x)z = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (X(t, x + \epsilon z) - X(t, x))$$

where $z \in H$. By using Galerkin approximations it is not difficult to show that the directional derivative $\eta^z(t, x)$ does exist and it is the solution of the equation,

$$\eta^z(t, x) = e^{tA}z + \sum_{k=1}^{\infty} \int_0^t e^{(t-s)A} [g'(X(s, x))\eta^z(s, x)e_k] d\beta_k(s). \quad (29)$$

Lemma 7. *There exists two positive constants a_1 and λ_1 such that*

$$\mathbb{E}(|\eta^z(t, x)|^2) \leq a_1 |z|^2 e^{\lambda_1 t} \quad t \geq 0, \quad x \in H. \quad (30)$$

Proof. We have

$$\mathbb{E}(|\eta^z(t, x)|^2) = |e^{tA}z|^2 + \sum_{k=1}^{\infty} \mathbb{E} \int_0^t |e^{(t-s)A} [g'(Y(s, x))\eta^z(s, x)e_k]|^2 ds.$$

Arguing as in the proof of Lemma 3 and taking into account (21), we see that,

$$\begin{aligned} \mathbb{E}(|\eta^z(t, x)|^2) &\leq |e^{tA}z|^2 + \frac{2}{\pi} \|g'\|_0^2 \int_0^t F(2\pi^2(t-s)) \mathbb{E}(|\eta^z(s, x)|^2) ds \\ &\leq |e^{tA}z|^2 + \frac{8\|g'\|_0^2}{\sqrt{2\pi^2}} \int_0^t (t-s)^{-1/2} e^{(t-s)/2} \mathbb{E}(|\eta^z(s, x)|^2) ds. \end{aligned}$$

By Lemma 1 it follows that

$$\mathbb{E}(|\eta^z(t, x)|^2) \leq |e^{tA}z|^2 e^{\frac{16\|g'\|_0^2}{\sqrt{2\pi^2}} t^{1/2}}, \quad t \geq 0$$

and the conclusion follows. $\square$

We want now to estimate $\zeta^z(t, x) := X_{xx}(t, x)(z, z)$ where

$$X_{xx}(t, x)(z, z) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (\eta^z(t, x + \epsilon z) - \eta^z(t, x))$$

and $z \in H$. Formally $\zeta^z(t, x)$ is the solution of the equation,

$$\begin{aligned} \zeta^z(t, x) &= \sum_{k=1}^{\infty} \int_0^t e^{(t-s)A} [g'(Y(s, x))\zeta^z(s, x)e_k] d\beta_k(s) \\ &\quad + \sum_{k=1}^{\infty} \int_0^t e^{(t-s)A} [g''(Y(s, x))(\eta^z(s, x))^2 e_k] d\beta_k(s). \end{aligned} \quad (31)$$

Here a problem arises since the term

$$g''(Y(s, x))(\eta^z(s, x))^2 e_k, \quad (32)$$

which appears in the second integral, belongs to $L^4(0, 1)$ and not to $L^2(0, 1)$ in general. For this reason we first need an estimate for

$$\mathbb{E}(|\eta^z(t, x)|_{L^4(0,1)}^4) = \mathbb{E}(|[\eta^z(t, x)]^2|_{L^2(0,1)}^2).$$

To get this estimate we shall proceed in two steps.

- (i) We shall estimate $\mathbb{E}(|\eta^z(t, x)|_{L^2(0,1)}^4)$.
- (ii) We shall estimate $\mathbb{E}(|(-A)^{1/8}\eta^z(t, x)|_{L^2(0,1)}^4)$.

Then we notice that, by the Sobolev embedding theorem, we have

$$D((-A)^{1/8}) \subset L^4(0, 1)$$

and so we end up with the required estimate for $\mathbb{E}(|\eta^z(t, x)|_{L^4(0,1)}^4)$.

Let us write (29) as

$$\eta^z(t, x) = e^{tA}z + \int_0^t \Phi(t-s)dW(s), \quad (33)$$

where

$$\Phi(t-s) = e^{(t-s)A}\sigma(s) = \sum_{k=1}^{\infty} e^{(t-s)A}[g'(X(s, x))\eta^z(s, x)e_k]. \quad (34)$$

We shall use the following Burkholder estimate, see [11].

$$\mathbb{E} \left[\left| \int_0^t \Phi(t-s)dW(s) \right|^4 \right] \leq c\mathbb{E} \left[\left(\int_0^t \|\Phi(t-s)\|_{HS}^2 ds \right)^2 \right], \quad (35)$$

where c is a given positive constant and

$$\begin{aligned} \|\Phi(t-s)\|_{HS}^2 &= \sum_{h,k=1}^{\infty} \langle e^{(t-s)A}[g'(X(s, x))\eta^z(s, x)e_k], e_h \rangle^2 \\ &= \sum_{h,k=1}^{\infty} e^{-2\pi^2 h^2(t-s)} \langle g'(X(s, x))\eta^z(s, x)e_k, e_h \rangle^2 \\ &= \sum_{h=1}^{\infty} e^{-2\pi^2 h^2(t-s)} |g'(X(s, x))\eta^z(s, x)e_h|^2 \\ &\leq \frac{2}{\pi} \|g\|_1^2 F(2\pi^2(t-s)) |\eta^z(s, x)|^2 \\ &\leq \frac{32}{\pi^2} \|g\|_1^2 (t-s)^{-1/2} |\eta^z(s, x)|^2. \end{aligned} \quad (36)$$

where we have used (21). Now we are ready to prove

Lemma 8. *Let $z \in L^2(0, 1)$. Then there exists constants $a_2 > 0$ and $\lambda_2 > 0$ such that*

$$\mathbb{E}(|\eta^z(t, x)|_{L^2(0,1)}^4) \leq a_2 e^{\lambda_2 t} |z|_{L^2(0,1)}^4, \quad t \geq 0, x \in H. \quad (37)$$

Proof. Let $z \in L^2(0, 1)$. By (33) we have

$$\mathbb{E}(|\eta^z(t, x)|_{L^2(0,1)}^4) \leq 8|e^{tA}z|_{L^2(0,1)}^4 + 8\mathbb{E}\left[\left|\int_0^t \Phi(t-s)dW(s)\right|_{L^2(0,1)}^4\right].$$

which, taking into account (35), yields

$$\mathbb{E}(|\eta^z(t, x)|_{L^2(0,1)}^4) \leq 8|e^{tA}z|_{L^2(0,1)}^4 + 8c\mathbb{E}\left[\left(\int_0^t \|\Phi(t-s)\|_{HS}^2 ds\right)^2\right]. \quad (38)$$

Let us estimate the term $J := \mathbb{E}\left[\left(\int_0^t \|\Phi(t-s)\|_{HS}^2 ds\right)^2\right]$. We have by (36)

$$\begin{aligned} J &\leq \frac{2^{10}}{\pi^4} \|g\|_1^4 \mathbb{E}\left[\left(\int_0^t (t-s)^{-1/2} |\eta^z(s, x)|_{L^2(0,1)}^2 ds\right)^2\right] \\ &= \frac{2^{10}}{\pi^4} \|g\|_1^4 \int_0^t \int_0^t (t-s)^{-1/2} (t-s_1)^{-1/2} \\ &\quad \mathbb{E}\left[|\eta^z(s, x)|_{L^2(0,1)}^2 |\eta^z(s_1, x)|_{L^2(0,1)}^2\right] ds ds_1 \\ &\leq \frac{2^9}{\pi^4} \|g\|_1^4 \int_0^t \int_0^t (t-s)^{-1/2} (t-s_1)^{-1/2} \mathbb{E}\left[|\eta^z(s, x)|_{L^2(0,1)}^4\right] ds ds_1 \\ &\quad + \frac{2^9}{\pi^4} \|g\|_1^4 \int_0^t \int_0^t (t-s)^{-1/2} (t-s_1)^{-1/2} \mathbb{E}\left[|\eta^z(s_1, x)|_{L^2(0,1)}^4\right] ds ds_1 \\ &\leq \frac{2^{11}}{\pi^4} \|g\|_1^4 t^{1/2} \int_0^t (t-s)^{-1/2} \mathbb{E}\left[|\eta^z(s, x)|_{L^2(0,1)}^4\right] ds. \end{aligned}$$

Substituting in (38) yields

$$\begin{aligned} \mathbb{E}(|\eta^z(t, x)|_{L^2(0,1)}^4) &\leq 8|e^{tA}z|_{L^2(0,1)}^4 \\ &\quad + c \frac{2^{14}}{\pi^4} \|g\|_1^4 t^{1/2} \int_0^t (t-s)^{-1/2} \mathbb{E}\left[|\eta^z(s, x)|_{L^2(0,1)}^4\right] ds. \end{aligned}$$

Now by the Gronwall Lemma it follows that there exist positive constants ρ, l such that

$$\mathbb{E}(|\eta^z(t, x)|_{L^2(0,1)}^4) \leq 8|e^{tA}z|_{L^2(0,1)}^4 + \rho \int_0^t (t-s)^{-1/2} e^{l(t-s)} |e^{sA}z|_{L^2(0,1)}^4 ds, \quad (39)$$

which implies the conclusion. $\square$

Lemma 9. *Let $z \in L^2(0, 1)$. Then there exists constants $a_3 > 0$ and $\lambda_3 > 0$ such that*

$$\mathbb{E}(|(-A)^{1/8}\eta^z(t, x)|_{L^2(0,1)}^4) \leq a_3 e^{\lambda_3 t} |(-A)^{1/8} e^{tA} z|_{L^2(0,1)}^4, \quad t \geq 0, x \in H. \quad (40)$$

Proof. Let $z \in L^2(0, 1)$. Then we have

$$\begin{aligned} \mathbb{E}(|(-A)^{1/8}\eta^z(t, x)|_{L^2(0,1)}^4) &\leq 8|(-A)^{1/8} e^{tA} z|_{L^2(0,1)}^4 \\ &+ 8\mathbb{E} \left[\left| \int_0^t \Phi_1(t-s) dW(s) \right|_{L^2(0,1)}^4 \right]. \end{aligned}$$

where

$$\Phi_1(t-s) = (-A)^{1/8} e^{(t-s)A} \sigma(s) = \sum_{k=1}^{\infty} (-A)^{1/8} e^{(t-s)A} [g'(X(s, x)) \eta^z(s, x) e_k].$$

We have

$$\begin{aligned} \|\Phi_1(t-s)\|_{HS}^2 &= \sum_{h,k=1}^{\infty} \langle (-A)^{1/8} e^{(t-s)A} [g'(X(s, x)) \eta^z(s, x) e_k], e_h \rangle^2 \\ &= \sum_{h,k=1}^{\infty} (\pi h)^{1/2} e^{-2\pi^2 h^2 (t-s)} \langle g'(X(s, x)) \eta^z(s, x) e_k, e_h \rangle^2 \\ &= \sum_{h=1}^{\infty} (\pi h)^{1/2} e^{-2\pi^2 h^2 (t-s)} |g'(X(s, x)) \eta^z(s, x) e_h|^2. \end{aligned}$$

It is not difficult to show that there is a positive constant d_1 such that

$$\|\Phi_1(t-s)\|_{HS}^2 \leq d_1 (t-s)^{-3/4} |\eta^z(s, x)|^2. \quad (41)$$

Now we have

$$\begin{aligned} \mathbb{E}(|(-A)^{1/8}\eta^z(t, x)|_{L^2(0,1)}^4) &\leq 8|(-A)^{1/8} e^{tA} z|_{L^2(0,1)}^4 \\ &+ 8c\mathbb{E} \left[\left(\int_0^t \|\Phi_1(t-s)\|_{HS}^2 ds \right)^2 \right]. \end{aligned} \quad (42)$$

By proceeding as in the proof of the previous lemma we find that there is $d_2 > 0$ such that

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^t \|\Phi_1(t-s)\|_{HS}^2 ds \right)^2 \right] \\ \leq d_2 \|g\|_1^4 t^{1/4} \int_0^t (t-s)^{-3/4} \mathbb{E} \left[|\eta^z(s, x)|_{L^2(0,1)}^4 \right] ds \end{aligned}$$

which yields

$$\begin{aligned} \mathbb{E}(|(-A)^{1/8}\eta^z(t, x)|_{L^2(0,1)}^4) &\leq 8|(-A)^{1/8}e^{tA}z|_{L^2(0,1)}^4 \\ &\quad + d_2 t^{1/2} \int_0^t (t-s)^{-3/4} \mathbb{E} \left[|\eta^z(s, x)|_{L^2(0,1)}^4 \right] ds. \end{aligned}$$

So, the conclusion follows from Lemma 8. $\square$

Now, using the Sobolev embedding $D((-A)^{1/8}) \subset L^4(0, 1)$ we can conclude that

Lemma 10. *Let $z \in L^2(0, 1)$. Then there exists constants $a_4 > 0$ and $\lambda_4 > 0$ such that*

$$\mathbb{E}(|\eta^z(t, x)|_{L^4(0,1)}^4) \leq a_4 e^{\lambda_4 t} |z|_{L^2(0,1)}^4, \quad t \geq 0, x \in H. \quad (43)$$

Now we are in position to estimate ζ^z for $z \in L^2(0, 1)$.

Lemma 11. *There exists constants $a_5 > 0$ and $\lambda_5 > 0$ such that for all $z \in L^2(0, 1)$, we have*

$$\mathbb{E}(|\zeta^z(t, x)|^2) \leq a_5 |z|_{L^2(0,1)}^4 e^{\lambda_5 t}, \quad t \geq 0, x \in H. \quad (44)$$

Proof. Let $z \in L^2(0, 1)$. Using Galerkin approximations we can show that (31) holds. From (31) we deduce that,

$$\begin{aligned} \mathbb{E}(|\zeta^z(t, x)|^2) &\leq 2\mathbb{E} \sum_{k=1}^{\infty} \int_0^t |e^{(t-s)A} [g'(X(s, x)) \zeta^z(s, x) e_k]|^2 ds \\ &\quad + 2\mathbb{E} \sum_{k=1}^{\infty} \int_0^t |e^{(t-s)A} [g''(X(s, x)) (\eta^z(s, x))^2 e_k]|^2 ds : \\ &= 2J_1 + 2J_2. \end{aligned}$$

Arguing as in the proof of Lemma 3 we find

$$\begin{aligned} J_1 &= \mathbb{E} \sum_{h,k=1}^{\infty} |\langle e^{(t-s)A} [g'(X(s, x)) \zeta^z(s, x) e_k], e_h \rangle|^2 \\ &= \mathbb{E} \sum_{h,k=1}^{\infty} e^{-2(t-s)\pi^2 h^2} |\langle g'(X(s, x)) \zeta^z(s, x) e_k, e_h \rangle|^2 \\ &= \mathbb{E} \sum_{h=1}^{\infty} e^{-2(t-s)\pi^2 h^2} |g'(X(s, x)) \zeta^z(s, x) e_h|^2 \\ &\leq \frac{2}{\pi} \|g\|_1^2 F(2(t-s)\pi^2) \mathbb{E} |\zeta^z(s, x)|^2 \leq \frac{4}{\pi^2} (t-s)^{-1/2} \mathbb{E} |\zeta^z(s, x)|^2. \end{aligned} \quad (45)$$

Similarly for J_2 we find, taking into account Lemma 10,

$$\begin{aligned}
J_2 &= \mathbb{E} \sum_{h,k=1}^{\infty} |\langle e^{(t-s)A} [g''(X(s,x))(\eta^z(s,x))^2 e_k], e_h \rangle|^2 \\
&= \mathbb{E} \sum_{h,k=1}^{\infty} e^{-2(t-s)\pi^2 h^2} |\langle g'(X(s,x))(\eta^z(s,x))^2 e_k, e_h \rangle|^2 \\
&= \mathbb{E} \sum_{h=1}^{\infty} e^{-2(t-s)\pi^2 h^2} |g'(X(s,x))(\eta^z(s,x))^2 e_h|^2 \\
&\leq \frac{2}{\pi} \|g\|_1^2 F(2(t-s)\pi^2) \mathbb{E} |(\eta^z(s,x))^2|^2 \\
&\leq \frac{4}{\pi^2} (t-s)^{-1/2} \mathbb{E} |(\eta^z(s,x))^2|^2 \leq a_4 \frac{4}{\pi^2} (t-s)^{-1/2} e^{\lambda_4 s} |z|_{L^2(0,1)}^4.
\end{aligned} \tag{46}$$

Now the conclusion follows from (45), (46) and the Gronwall lemma. $\square$

Remark 12. By Lemma 11 it follows that if $\varphi \in C_b^2(H)$ the function $u(t, \cdot)$ possesses bounded second order derivatives in all directions of H for any $t \geq 0$. So, it is Frécher differentiable and belongs to $C_b^1(H)$ (more precisely to $C_b^{1+\varepsilon}(H)$ for all $\varepsilon \in (0, 1)$).

3.3 Strict solutions of the Kolmogorov equation

We are now in position to show existence of a strict solution $u(t, x)$ (in the sense that $u(t, x)$ fulfills conditions (i)-(iv) of Proposition 13 below) of equation (26) for all $\varphi \in C_b^2(H)$. Let us define,

$$\bar{\lambda} = \frac{1}{2} \max\{\lambda_i : i = 1, \dots, 5\}$$

and

$$\kappa = \frac{1}{2} \max\{a_i^{1/2} : i = 1, \dots, 5\}$$

Theorem 13. Assume that $\varphi \in C_b^2(H)$, $D^2\varphi(x)$ is of trace class for any $x \in H$ and $\text{Tr}[D^2\varphi] \in C_b(H)$. Let

$$u(t, x) = \mathbb{E}[\varphi(X(t, x))], \quad t \geq 0, x \in H,$$

where $X(t, x)$ is the mild solution of (17). Then the following statements hold.

- (i) For all $t \geq 0$, $u(t, \cdot) \in C_b^1(H)$ and possesses second order derivatives in all directions of H .

(ii) For all $t > 0$ and any $x \in H$ we have

$$|u_x(t, x)| \leq \kappa e^{\bar{\lambda}t} \|\varphi\|_1 \quad (47)$$

and

$$|u_{xx}(t, x)| \leq \kappa e^{\bar{\lambda}t} \|\varphi\|_2. \quad (48)$$

(iii) There exists $\kappa_1 > 0$ such that for all $t \geq 0$ and any $x \in H$ we have

$$\begin{aligned} |\text{Tr} [\sigma^2(x) u_{xx}(t, x)]| &= \left| \sum_{k=1}^{\infty} \langle u_{xx}(t, x) (\sigma(x) e_k, \sigma(x) e_k) \rangle \right| \\ &\leq \kappa_1 e^{\bar{\lambda}t} \|\varphi\|_2 (1 + \sup_{x \in H} \|D^2 \varphi(x)\|_{L_1(H)}). \end{aligned} \quad (49)$$

(iv) For all $x \in D(A)$, $u(\cdot, x)$ is differentiable in $(0, +\infty)$ and fulfills (26).

Proof. Let us prove (i). For any $x, z \in H$ and $t \geq 0$ we have

$$\langle u_x(t, x), z \rangle = \mathbb{E}[\langle D\varphi(X(t, x)), \eta^z(t, x) \rangle]$$

Therefore

$$|\langle u_x(t, x), z \rangle| \leq \|\varphi\|_1 |\eta^z(t, x)|.$$

By Lemma 7 it follows that

$$|\langle u_x(t, x), z \rangle| \leq \|\varphi\|_1 a_1^{1/2} e^{\lambda_1 t/2}.$$

Thus (47) follows from the arbitrariness of z . Moreover $u(t, \cdot) \in C_b^1(H)$ in view of Remark 12.

Let us prove (ii). For any $x, z \in H$ and $t \geq 0$ we have

$$\langle u_{xx}(t, x) z, z \rangle = \mathbb{E}[\langle D\varphi(X(t, x)), \zeta^z(t, x) \rangle] + \mathbb{E}[\langle D^2 \varphi(X(t, x)) \eta^z(t, x), \eta^z(t, x) \rangle].$$

Therefore

$$\begin{aligned} |\langle u_{xx}(t, x) z, z \rangle| &\leq \|\varphi\|_1 \mathbb{E}[|\zeta^z(t, x)|] + \|\varphi\|_2 \mathbb{E}[|\eta^z(t, x)|^2] \\ &\leq \|\varphi\|_1 \sqrt{a_5} e^{\lambda_5 t/2} |z|^2 + \|\varphi\|_2 a_1 e^{\lambda_1 t/2} |z|^2 \end{aligned}$$

and (ii) follows.

Let us prove (iii). For any $x, z \in H$ and $t \geq 0$ we have

$$\begin{aligned} \text{Tr} [\sigma^2(x) u_{xx}(t, x)] &= \sum_{k=1}^{\infty} \langle u_{xx}(t, x) (\sigma(x) e_k, \sigma(x) e_k) \rangle \\ &= \sum_{k=1}^{\infty} \mathbb{E}[\langle D\varphi(X(t, x)), \zeta^{\sigma(x) e_k}(t, x) \rangle] \\ &\quad + \text{Tr} \mathbb{E}[\sigma^2(x) X_x(t, x) D^2 \varphi(X(t, x)) X_x^*(t, x)] \\ &:= J_1(t, x) + J_2(t, x). \end{aligned}$$

Therefore

$$|J_2(t, x)| \leq a_1 e^{\lambda_1 t} \|D^2 \varphi\|_{HS}.$$

Concerning J_1 we have

$$|J_1(t, x)| \leq \|\varphi\|_1 |T(t, x)|$$

where

$$T(t, x) = \sum_{k=1}^{\infty} \zeta^{\sigma(x)e_k}(t, x).$$

Then one can check that $T(t, x)$ is the solution of the equation

$$\begin{aligned} T(t, x) = & \int_0^t e^{(t-s)A} [g'(X(s, x)) T(s, x)] dW(s) \\ & + \int_0^t e^{(t-s)A} [g''(X(s, x)) K(s, x)] dW(s), \end{aligned} \quad (50)$$

where

$$K(t, x) = \sum_{k=1}^{\infty} (\eta^{\sigma(x)e_k}(t, x))^2.$$

Using estimate (39), it is not difficult to show that equation (50) has a solution and estimate (49) holds.

Let us prove finally (iv). Assume that $x \in D(A)$ and let $u^n(t, x)$ be the solution of (28). Then, taking into account that estimates from Lemmas 7 and 11 can also be proved for the function u^n with constants independent of n , it is not difficult to check that,

$$\lim_{n \rightarrow \infty} u^n(t, x) = u(t, x), \quad t > 0, x \in H,$$

$$\lim_{n \rightarrow \infty} u_x^n(t, x) = u_x(t, x), \quad t > 0, x \in H,$$

and

$$\lim_{n \rightarrow \infty} \text{Tr} [\sigma(x) C \sigma(x) u_{xx}^n(t, x)] = \text{Tr} [\sigma(x) C \sigma(x) u_{xx}(t, x)], \quad t > 0, x \in H.$$

Consequently,

$$\lim_{n \rightarrow \infty} u_t^n(t, x) = u_t(t, x), \quad t > 0, x \in D(A),$$

and the conclusion follows. $\square$

We consider finally the elliptic Kolmogorov equation

$$\lambda \varphi(x) - \frac{1}{2} \text{Tr} [\sigma^2(x) \varphi_{xx}(x)] - \langle Ax, \varphi_x(x) \rangle = f(x), \quad x \in D(A), \quad (51)$$

where $\lambda > 0$ and $f \in C_b(H)$ are given.

Theorem 14. Assume that $\lambda > \bar{\lambda}$, $f \in C_b^2(H)$, $D^2f(x)$ is of trace class for any $x \in H$ and $\text{Tr} [D^2f] \in C_b(H)$. Define

$$\varphi(x) = \int_0^\infty e^{-\lambda t} \mathbb{E}[f(X(t, x))] dt, \quad t \geq 0, x \in H,$$

Then the following statements hold.

- (i) $\varphi \in C_b^1(H)$ and possesses second order derivatives in all directions of H .
- (ii) For all $x \in H$ we have

$$|u_x(t, x)| \leq \frac{\kappa}{\lambda - \bar{\lambda}} \|\varphi\|_1 \quad (52)$$

and

$$|u_{xx}(t, x)| \leq \frac{\kappa}{\lambda - \bar{\lambda}} \|\varphi\|_2. \quad (53)$$

- (iii) There exists $\kappa_1 > 0$ such that for all $x \in H$ we have

$$\begin{aligned} |\text{Tr} [\sigma^2(x) u_{xx}(t, x)]| &= \left| \sum_{k=1}^\infty \langle u_{xx}(t, x) (\sigma(x) e_k, \sigma(x) e_k) \rangle \right| \\ &\leq \frac{\kappa_1}{\lambda - \bar{\lambda}} \|\varphi\|_2 (1 + \|D^2\varphi\|_{HS}). \end{aligned} \quad (54)$$

- (ii) We have

$$|\varphi_x(x)| \leq \frac{\sqrt{a_1}}{\lambda - \bar{\lambda}} \|f\|_1, \quad x \in H, \quad (55)$$

and

$$|\text{Tr} [\sigma^2(x) \varphi_{xx}(t, x)]| \leq \frac{\kappa}{(\lambda - \bar{l})} \|f\|_2, \quad x \in H. \quad (56)$$

- (iv) For all $x \in D(A)$ the equation (51) is fulfilled.

Proof. The conclusion follows from Proposition 13 and estimates (47), (48) and (49). $\square$

3.4 The Kolmogorov operator

It is well known that the semigroup R_t is not in general strongly continuous in $C_b(H)$. However, we can define its infinitesimal generator by proceeding as in [4]. Namely, for any $\lambda > 0$ and any $f \in C_b(H)$ we define

$$F_\lambda(f)(x) = \int_0^\infty e^{-\lambda t} R_t f(x) dt, \quad x \in H.$$

Proposition 15. *For any $f \in C_b(H)$ and any $\lambda > 0$ we have $F_\lambda(f) \in C_b(H)$ and the following estimate holds*

$$\|F_\lambda(f)\|_0 \leq \frac{1}{\lambda} \|f\|_0. \quad (57)$$

Moreover there exists a unique closed operator $L: D(L) \subset C_b(H) \rightarrow C_b(H)$ such that for any $\lambda > 0$ and any $f \in C_b(H)$ we have $F_\lambda(f) = R(\lambda, L)f$.

Proof. Let first $f \in C_b^1(H)$; then it is obvious that if $F_\lambda(f) \in C_b(H)$ the inequality (57) holds. Moreover for all $x, y \in H$ we have

$$\begin{aligned} |F_\lambda(f)(x) - F_\lambda(f)(y)| &\leq \int_0^\infty e^{-\lambda t} \mathbb{E}(|f(X(t, x)) - f(X(t, y))|) dt \\ &\leq \|f\|_1 \int_0^\infty e^{-\lambda t} \mathbb{E}|X(t, x) - X(t, y)| dt. \end{aligned} \quad (58)$$

On the other hand we have

$$X(t, x) - X(t, y) = \int_0^1 X_x(t, (1-r)x + ry)(x - y) dr,$$

so that, recalling Lemma 7, we find

$$|X(t, x) - X(t, y)| \leq \sqrt{a_1} e^{\frac{1}{2} \lambda_1 t} |x - y|. \quad (59)$$

Now, substituting this inequality in (58) yields

$$|F_\lambda(f)(x) - F_\lambda(f)(y)| \leq \sqrt{a_1} \int_0^\infty e^{(\frac{1}{2} \lambda_1 - \lambda)t} dt |x - y|.$$

Thus, if $\lambda > \frac{1}{2} \lambda_1$ we have proved that $F_\lambda(f) \in C_b(H)$ (it is even Lipschitz). Since $C_b^1(H)$ is dense in $C_b(H)$ we can conclude that $F_\lambda(f) \in C_b(H)$ for all $f \in C_b(H)$ (and $\lambda > \frac{1}{2} \lambda_1$).

Now it is easy to see that F_λ fulfills the resolvent identity

$$F_\lambda - F_\mu = (\mu - \lambda)F_\lambda F_\mu, \quad \lambda, \mu > 0.$$

So, by a classical result, see e. g. [18], there exists a unique closed operator $L: D(L) \subset C_b(H) \rightarrow C_b(H)$ such that for any $\lambda > \frac{1}{2} \lambda_1$ and any $f \in C_b(H)$ we have $F_\lambda(f) = R(\lambda, L)f$.

Finally, by (57) we see that L is m -dissipative so that condition $\lambda > \frac{1}{2} \lambda_1$ can be replaced by $\lambda > 0$. $\square$

Remark 16. Assume that $f \in C_b^2(H)$, $D^2 f(x)$ is of trace class for any $x \in H$ and $\text{Tr} [D^2 f] \in C_b(H)$. Let moreover $\lambda > 0$ and $\varphi = R(\lambda, L)$. Then by Proposition 14 it follows that

$$L\varphi(x) = \frac{1}{2} \text{Tr} [\sigma^2(x) \varphi_{xx}(x)] + \langle Ax, \varphi_x(x) \rangle, \quad x \in D(A). \quad (60)$$

Now we are going to prove, following an argument in [10], that if $\varphi \in R(\lambda, L)(C_b^2(H))$ we have $\varphi^2 \in D(L)$.

Proposition 17. *Assume that $f \in C_b^2(H)$, $D^2f(x)$ is of trace class for any $x \in H$ and $\text{Tr}[D^2f] \in C_b(H)$. Let moreover $\lambda > 0$ and $\varphi = R(\lambda, L)$. Then $\varphi^2 \in D(L)$ and*

$$L(\varphi^2) = 2\varphi L\varphi + |\sigma D\varphi|^2. \quad (61)$$

Proof. Let L_n be the approximating Kolmogorov operator

$$L_n\varphi(x) = \frac{1}{2} \text{Tr}[\sigma^2(x)P_n\varphi_{xx}(x)] + \langle A_n x, \varphi_x(x) \rangle, \quad \varphi \in C_b(H), x \in H \quad (62)$$

and let $\varphi^n = R(\lambda, L_n)f$. Then, by a straightforward computation, it follows that

$$L_n((\varphi^n)^2) = 2\varphi^n L_n\varphi^n + |\sigma P_n D\varphi^n|^2.$$

Now, multiplying both sides of the equation $\lambda\varphi^n - L_n\varphi^n = f$ by φ^n , yields

$$\lambda(\varphi^n)^2 - L_n\varphi^n \varphi^n = f\varphi^n,$$

which is equivalent to

$$2\lambda(\varphi^n)^2 - L_n((\varphi^n)^2) = 2f\varphi^n - |\sigma P_n D\varphi^n|^2.$$

Therefore,

$$(\varphi^n)^2 = R(2\lambda, L_n)(2f\varphi^n - |\sigma P_n D\varphi^n|^2).$$

Letting $n \rightarrow \infty$ yields

$$\varphi^2 = R(2\lambda, L)(2f\varphi - |\sigma D\varphi|^2).$$

Consequently

$$2\lambda\varphi^2 - L\varphi^2 = 2f\varphi - |\sigma(x)D\varphi|^2,$$

which yields (61). $\square$

4 Invariant measures

4.1 Existence and uniqueness

We denote by $\mathcal{P}(H)$ the set of all Borel probability measures on H . We recall that a probability measure $\mu \in \mathcal{P}(H)$ is said to be *invariant* for the transition semigroup R_t defined by (7) if

$$\int_H R_t\varphi d\mu = \int_H \varphi d\mu \quad \text{for all } \varphi \in C_b(H). \quad (63)$$

Theorem 18. *There is an invariant measure μ for R_t . Moreover, for any $\beta \in [0, 1/4)$ we have*

$$\int_H |(-A)^\beta x|^2 \mu(dx) < +\infty. \quad (64)$$

Finally, if $1/g$ is bounded the invariant measure μ is unique.

Proof. Let $X(t, x)$ be the solution of (18). Using Lemma 3 and inequality (21), we find that

$$\begin{aligned} \mathbb{E}(|X(t, x)|^2) &\leq 2e^{-2\pi^2 t} |x|^2 + 2 \sum_{h=1}^{\infty} \int_0^t e^{-2\pi^2 h^2(t-s)} \mathbb{E}(|\sigma(X(s, x))e_h|^2) ds \\ &\leq 2e^{-2\pi^2 t} |x|^2 + \|g\|_0^2 \int_0^t F(2\pi^2(t-s)) ds \\ &\leq 2e^{-2\pi^2 t} |x|^2 + 4\|g\|_0^2 \int_0^t e^{-\pi^2(t-s)} (2\pi^2(t-s))^{-1/2} ds. \end{aligned}$$

So,

$$\mathbb{E}(|X(t, x)|^2) \leq 2e^{-2\pi^2 t} |x|^2 + \frac{4}{\sqrt{\pi}} \|g\|_0^2. \quad (65)$$

Now let $\beta \in (0, 1/4)$. Using the well known estimate

$$\|(-A)^\beta e^{tA}\| \leq c_\beta t^{-\beta} e^{-\pi^2 t}, \quad t \geq 0, \quad (66)$$

where c_β is a suitable constants, we find,

$$\begin{aligned} &\mathbb{E}(|(-A)^\beta X(t, x)|^2) \\ &\leq 2c_\beta t^{-2\beta} e^{-2\pi^2 t} |x|^2 + 2 \int_0^t \sum_{h=1}^{\infty} (\pi h)^{4\beta} e^{-2\pi^2 h^2(t-s)} \mathbb{E}(|\sigma(X(s, x))e_h|^2) ds \\ &\leq 2c_\beta t^{-2\beta} e^{-2\pi^2 t} |x|^2 + 2\|g\|_0^2 \int_0^t F_\beta(2\pi^2(t-s)) ds \end{aligned}$$

where F_β is defined by

$$F_\beta(t) = \sum_{h=1}^{\infty} h^{4\beta} e^{-h^2 t}, \quad t > 0.$$

It is not difficult to show that there is $k_\beta > 0$ such that

$$F_\beta(t) \leq k_\beta t^{-1/2-2\beta} e^{-t}, \quad t \geq 0.$$

Now we have

$$\begin{aligned}
\mathbb{E}(|(-A)^\beta X(t, x)|^2) &\leq 2c_\beta t^{-2\beta} e^{-2\pi^2 t} |x|^2 \\
&+ 2\|g\|_0^2 k_\beta \int_0^t (2\pi^2 s)^{-1/2-2\beta} e^{-2\pi^2 s} ds \\
&\leq 2c_\beta t^{-2\beta} e^{-2\pi^2 t} |x|^2 + c_\beta k_\beta \|g\|_0^2.
\end{aligned} \tag{67}$$

Since the embedding $D(A) \subset H$ is compact, the existence of an invariant measure follows from the Krylov–Bogoliubov theorem.

Let us show (64). Let $\gamma > 0$ and set

$$\varphi_\gamma(x) = \frac{|x|^2}{1 + \gamma|x|^2}, \quad x \in H.$$

Then $\varphi_\gamma \in C_b(H)$ and, proceeding as in the proof of (65) we see that there exists a constant $\kappa > 0$ (independent on λ) such that,

$$R_t(\varphi_\gamma)(x) = \mathbb{E}[\varphi_\gamma(X(t, x))] \leq e^{-2\pi^2 t} \varphi_\gamma(x) + \kappa. \tag{68}$$

Integrating both sides of (68) with respect to x over H and taking into account the invariance of μ yields,

$$\int_H \varphi_\gamma(x) \mu(dx) \leq e^{-2\pi^2 t} \int_H \varphi_\gamma(x) \mu(dx) + \kappa.$$

Therefore, there exists $\kappa_1 > 0$ (independent on λ) such that

$$\int_H \varphi_\gamma(x) \mu(dx) \leq \kappa_1.$$

Letting γ tend to 0 yields,

$$\int_H |x|^2 \mu(dx) \leq \kappa_1. \tag{69}$$

Now, integrating both sides of (67) with respect to x over H and taking into account again the invariance of μ yields,

$$\int_H |(-A)^{-\beta} x|^2 \mu(dx) \leq 2c_\beta t^{-2\beta} e^{-2\pi^2 t} \kappa_1 + c_\beta \|g\|_0^2,$$

and so, (64) follows. Finally, if $1/g$ is bounded the uniqueness of μ follows from the Doob Theorem since R_t is irreducible and strong Feller by [16].
□

Remark 19. More general results of existence of invariant measures can be found in the paper [7].

4.2 Existence of a core of smooth functions for L_μ

Let us fix an invariant measure μ for R_t . It is well known that R_t can be uniquely extended to a strongly continuous semigroup of contractions on $L^2(H, \mu)$ which we shall still denote by R_t . The infinitesimal generator of R_t in $L^2(H, \mu)$ will be denoted by L_μ . Since R_t is a contraction semigroup, L_μ is m -dissipative in $L^2(H, \mu)$.

In this subsection we want to define a core of L_μ consisting of regular functions.

Proposition 20. *Set*

$$\Lambda = \{\varphi \in C_b^2(H) : D^2\varphi(x) \in L_1(H) \text{ for all } x \in H \text{ and } \text{Tr} [D^2\varphi] \in C_b(H)\}$$

and

$$\Gamma := \bigcup_{\lambda > 0} R(\lambda, L)(\Lambda).$$

Then Γ is a core for L_μ .

Moreover if $\varphi \in \Gamma$ we have $\varphi^2 \in D(L_\mu)$ and the following identity holds

$$L_\mu(\varphi^2) = 2\varphi L_\mu\varphi + |\sigma D\varphi|^2. \quad (70)$$

Proof. Let $\lambda > 0$. It is clear that any $\varphi \in D(L)$ belongs to $D(L_\mu)$ as well, so that we have

$$(\lambda - L_\mu)(\Gamma) = (\lambda - L)(\Gamma) \supset \Lambda.$$

Since Λ is dense in $L^2(H, \mu)$ (by a standard argument of monotone classes), we can conclude that $(\lambda - L_\mu)(\Gamma)$ is dense in $L^2(H, \mu)$. Now the Lumer-Phillips theorem implies that Γ is a core for L_μ .

Finally, the last statement follows from (61). $\square$

5 The basic integration by parts formula

In this section we assume that g^{-1} is bounded. We recall that in this case μ is the unique invariant measure of the transition semigroup R_t .

Proposition 21. *The operator*

$$D : \Gamma \subset L^2(H, \mu) \rightarrow L^2(H, \mu; H), \quad \varphi \mapsto D\varphi, \quad (71)$$

is uniquely extendible to a linear bounded operator $D : D(L_\mu) \rightarrow L^2(H, \mu; H)$, where $D(L_\mu)$ is endowed with the graph norm of L_μ . Moreover, the following identity holds

$$\int_H L_\mu\varphi \varphi \, d\mu = -\frac{1}{2} \int_H |\sigma D\varphi|^2 d\mu, \quad \varphi \in D(L_\mu). \quad (72)$$

Identity (72) is called in French “identité du carré du champs”. It will play an important rôle in what follows.

Proof. Let $\varphi \in \Gamma$, then $\varphi^2 \in D(L_\mu)$ in view of Proposition 20 we have

$$L_\mu \varphi^2 = 2\varphi L_\mu \varphi + |gD\varphi|^2.$$

Integrating this identity with respect to μ over H and taking into account that

$$\int_H L_\mu \varphi^2 d\mu = 0$$

by the invariance of μ , implies (72) when $\varphi \in \Gamma$. Let now $\varphi \in D(L_\mu)$. Since Γ is a core for L_μ , there exists a sequence $\{\varphi_n\} \subset \Gamma$ such that

$$\varphi_n \rightarrow \varphi, \quad L_0 \varphi_n \rightarrow L_\mu \varphi \quad \text{in } L^2(H, \mu).$$

By (72) it follows that

$$\int_H |\sigma D(\varphi_n - \varphi_m)|^2 d\mu = -2 \int_H L_\mu(\varphi_n - \varphi_m) (\varphi_n - \varphi_m) d\mu.$$

So, the sequence $\{\sigma D\varphi_n\}$ is Cauchy in $L^2(H, \mu; H)$ and the conclusion follows. $\square$

Proposition 22. *Let $\varphi \in L^2(H, \mu)$ and $t \geq 0$. Then, for any $T > 0$, the linear operator*

$$\sigma DR_t : D(L_\mu) \subset L^2(H, \mu) \rightarrow L^2(0, T; L^2(H, \mu; H)), \quad \varphi \rightarrow \sigma DR_t \varphi,$$

is uniquely extendible to a linear bounded operator, still denoted by σDR_t , from $L^2(H, \mu)$ into $L^2(0, T; L^2(H, \mu; H))$. Moreover the following identity holds

$$\int_H (R_t \varphi)^2 d\mu + \int_0^t ds \int_H |\sigma DR_s \varphi|^2 d\mu = \int_H \varphi^2 d\mu. \quad (73)$$

Proof. We first establish (73) for $\varphi \in D(L_\mu)$. In this case we have

$$\frac{d}{dt} R_t \varphi = L_\mu \varphi.$$

Multiplying scalarly this identity by $R_t \varphi$, integrating with respect to μ over H and using (73), yields,

$$\frac{d}{dt} \int_H (R_t \varphi)^2 d\mu + \int_H |\sigma DR_s \varphi|^2 d\mu = 0. \quad (74)$$

Now (73) follows integrating (74) with respect to t . The case when $\varphi \in L^2(H, \mu)$ can be handled by approximating φ by elements of $D(L_\mu)$. $\square$

5.1 The Sobolev space $W^{1,2}(H, \mu)$

To define the Sobolev space we first show that the mapping

$$D_\mu : \Gamma \subset L^2(H, \mu) \rightarrow L^2(H, \mu; H), \quad \varphi \rightarrow D_\mu \varphi \quad (75)$$

is closable. Notice the difference between the map D defined by (71) and the map D_μ . The first one is a bounded operator in $D(L_\mu)$ (endowed with the graph norm of L_μ) whereas the second will be a closable operator in $L^2(H, \mu)$.

To prove closability of D_μ we recall the following estimate from Lemma 7.

$$\mathbb{E}(|X_x(t, x)h|^2) \leq a_1 e^{-2\omega t} |h|^2 \quad t \geq 0, \quad h, x \in H, \quad (76)$$

where $\omega = \pi^2 - 8\pi^{-2} \|g'\|_0^2$.

Lemma 23. *Let $\{\varphi_n\} \subset \Gamma$ and let $G \in L^2(H, \mu; H)$ be such that*

$$\lim_{n \rightarrow \infty} D\varphi_n = G \quad \text{in } L^2(H, \mu; H).$$

Then, for any $t \geq 0$ we have

$$\lim_{n \rightarrow \infty} DR_t \varphi_n = \mathbb{E}[X_x^*(t, x)G(X(t, x))] \quad \text{in } L^2(H, \mu; H).$$

In particular, if $D\varphi_n \rightarrow 0$ in $L^2(H, \mu; H)$ we have $DR_t \varphi_n \rightarrow 0$ in $L^2(H, \mu; H)$ for all $t > 0$.

Proof. Write

$$DR_t \varphi_n(x) = E[X_x^*(t, x)D\varphi_n(X(t, x))], \quad t \geq 0, \quad x \in H.$$

Taking into account estimate (76) and the invariance of μ , yields

$$\begin{aligned} & \int_H |DR_t \varphi_n(x) - \mathbb{E}[X_x^*(t, x)G(X(t, x))]|^2 \mu(dx) \\ &= \int_H |\mathbb{E}[X_x^*(t, x)(D\varphi_n(X(t, x)) - G(X(t, x))]|^2 \mu(dx) \\ &\leq a_1 e^{-2\omega t} \int_H \mathbb{E}[|D\varphi_n(X(t, x)) - G(X(t, x))|^2] \mu(dx) \\ &= a_1 e^{-2\omega t} \int_H R_t(|D\varphi_n - G|^2)(x) \mu(dx) \\ &= a_1 e^{-2\omega t} \int_H |D\varphi_n(x) - G(x)|^2 \mu(dx). \end{aligned}$$

The conclusion of the lemma follows. $\square$

Proposition 24. D_μ is closable. Moreover, if φ belongs to the domain of the closure $\overline{D_\mu}$ of D_μ and $\overline{D_\mu}\varphi = 0$ we have that $\overline{D_\mu}R_t\varphi = 0$ for any $t > 0$.

Proof. Let $\{\varphi_n\} \subset \Gamma$ and $G \in L^2(H, \mu; H)$ be such that

$$\varphi_n \rightarrow 0 \quad \text{in } L^2(H, \mu), \quad D\varphi_n \rightarrow G \quad \text{in } L^2(H, \mu; H).$$

By (73) we have that

$$\int_H (R_t\varphi_n)^2 d\mu + \int_0^t ds \int_H |\sigma DR_s\varphi_n|^2 d\mu = \int_H \varphi_n^2 d\mu.$$

Letting $n \rightarrow \infty$ and taking into account that g is bounded below, yields

$$\lim_{n \rightarrow \infty} \int_0^t ds \int_H |DR_s\varphi_n|^2 d\mu = 0.$$

Consequently, by Lemma 23, it follows that

$$\int_0^t ds \int_H (\mathbb{E}[X_x^*(s, x)G(X(s, x))]^2 \mu(dx) = 0, \quad h \in H.$$

Then for almost all $t \geq 0$ we have that

$$\mathbb{E}[X_x^*(t, x)G(X(t, x))] = 0. \tag{77}$$

Now fix $h \in H$. Then we have,

$$\begin{aligned} & |\mathbb{E}[\langle G(X(t, x)), h \rangle]| \\ & \leq |\mathbb{E}[\langle G(X(t, x)), X_x(t, x) \cdot h \rangle]| + |\mathbb{E}[\langle G(X(t, x)), h - X_x(t, x) \cdot h \rangle]| \\ & = |\mathbb{E}[\langle G(X(t, x)), h - X_x(t, x) \cdot h \rangle]|. \end{aligned}$$

Taking into account the invariance of μ and (77), we find that

$$\begin{aligned} & \int_H |R_t(\langle G(x), h \rangle)| \mu(dx) = \int_H |\mathbb{E}[\langle G(X(t, x)), h \rangle]| \mu(dx) \\ & = \int_H |\mathbb{E}[\langle G(X(t, x)), h - X_x(t, x) \cdot h \rangle]| \mu(dx) \\ & \leq \left[\int_H \mathbb{E}[|G(X(t, x))|^2] \mu(dx) \right]^{1/2} \left[\int_H \mathbb{E}[|h - X_x(t, x) \cdot h|^2] \mu(dx) \right]^{1/2}. \end{aligned}$$

Therefore, as $t \rightarrow 0$ we find by the strong continuity of R_t in $L^1(H, \mu)$

$$\int_H |\langle G(x), h \rangle| \mu(dx) = 0$$

and by the arbitrariness of h it follows that $G = 0$ as required. Finally, the last statement follows from by Lemma 23. $\square$

By Proposition 24 it follows that the mapping

$$D_\mu : \Gamma \subset L^2(H, \mu) \rightarrow L^2(H, \mu; H), \quad \varphi \mapsto D\varphi,$$

is closable, let $\overline{D_\mu}$ its closure. We shall denote by $W^{1,2}(H, \mu)$ the domain of $\overline{D_\mu}$ and, if there is not possibility of confusion, we shall set $\overline{D_\mu} = D$.

Proposition 25. *We have $D(L_\mu) \subset W^{1,2}(H, \mu)$ with continuous embedding. Moreover, the following identity holds*

$$\int_H L_\mu \varphi \varphi \, d\mu = -\frac{1}{2} \int_H |\sigma D\varphi|^2 d\mu, \quad \varphi \in D(L_\mu). \quad (78)$$

Proof. Let $\varphi \in D(L_\mu)$. Since Γ is a core for L_μ , there exists a sequence $\{\varphi_n\} \subset \Gamma$ such that

$$\varphi_n \rightarrow \varphi, \quad L_0 \varphi_n \rightarrow L_\mu \varphi \quad \text{in } L^2(H, \mu).$$

By (72) it follows that

$$\int_H |\sigma D(\varphi_n - \varphi_m)|^2 d\mu \leq 2 \int_H |L_0(\varphi_n - \varphi_m)| |\varphi_n - \varphi_m| \, d\mu.$$

Therefore the sequence $(D\varphi_n)$ is Cauchy in $L^2(H, \mu; H)$. Since D is closed it follows that $\varphi \in W^{1,2}(H, \mu)$ as required. $\square$

5.2 The Poincaré inequality

Since $1/g$ is bounded, by Theorem 18 there is a unique invariant measure μ for R_t and by the Doob theorem, see e.g. [12], we have that,

$$\lim_{n \rightarrow \infty} R_t \varphi(x) = \int_H \varphi(y) \mu(dy), \quad x \in H, \quad (79)$$

for all $\varphi \in C_b(H)$.

Let us prove now the Poincaré inequality.

Proposition 26. *Assume that $\|g'\|_0^2 \leq \frac{1}{8} \pi^4$. Then, for any $\varphi \in W^{1,2}(H, \nu)$ we have*

$$\int_H |\varphi - \bar{\varphi}|^2 d\mu \leq \frac{a_1}{\omega} \|g\|_0^2 \|1/g\|_0^2 \int_H |g(x) D\varphi|^2 d\mu, \quad (80)$$

where $\bar{\varphi} = \int_H \varphi d\mu$ and $\omega = \pi^2 - 8\pi^{-2} \|g'\|_0^2$.

Proof. Let first $\varphi \in \Gamma$. Then by (73) we have

$$\int_H |R_t \varphi(x) - \bar{\varphi}|^2 \mu(dx) = \int_0^t ds \int_H |\sigma D R_s \varphi|^2 d\mu. \quad (81)$$

Moreover by (76) it follows that

$$\begin{aligned} \mathbb{E}[|D R_s \varphi(x)|^2] &\leq \mathbb{E}[|D \varphi(X(s, x))|^2 |X_x(s, x)|^2] \\ &\leq a_1 e^{-2\omega s} \mathbb{E}[|D \varphi(X(s, x))|^2] = a_1 e^{-2\omega s} R_s(|D \varphi|^2)(x). \end{aligned}$$

Taking into account (79) and the invariance of μ we obtain,

$$\begin{aligned} &\int_H |R_t \varphi(x) - \bar{\varphi}|^2 \mu(dx) \\ &\leq a_1 \|g\|_0^2 \int_0^{+\infty} e^{-2\omega s} ds \int_H R_s(|D \varphi|^2)(x) \mu(dx) \\ &\leq a_1 \|g\|_0^2 \|1/g\|_0^2 \int_0^{+\infty} e^{-2\omega s} ds \int_H |g(x) D \varphi(x)|^2 \mu(dx), \end{aligned}$$

and the conclusion follows. If $\varphi \in W^{1,2}(H, \mu)$, we proceed by density. $\square$

Remark 27. If g is constant the condition $\|g'\|_0^2 \leq \frac{1}{8} \pi^4$ is trivially fulfilled and we recover a result in [10].

Remark 28. It is well known, see e.g. [8], that the Poincaré inequality implies that the spectrum $\sigma(L_\mu)$ of L_μ consists of 0 and a set included in the half-space

$$\{\lambda \in \mathbb{C} : \Re \lambda \leq -\omega_1\},$$

(spectral gap) where ω_1 is a positive constant.

The spectral gap in turn implies an exponential convergence of $R_t \varphi$ to the equilibrium

$$\int_{\mathbb{R}} |R_t \varphi - \bar{\varphi}|^2 d\nu \leq c e^{-2\omega_1 t} \int_{\mathbb{R}} |\varphi|^2 d\nu, \quad \varphi \in L^2(\mathbb{R}, \nu), \quad (82)$$

where c is a suitable constant.

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